Bivalve Meat Dry Weight: A Farm Protocol

Oct 23, 2026 · The Nature Conservancy

Measures dry tissue weight (DTW), grams per animal

What this protocol gives you

This protocol provides the average dry meat weight of one animal, in grams. The Filtration Service Estimator calls this **dry tissue weight (DTW)**. Here, "meat" means all of the soft body inside the shell, not just the part people eat. Please note that the instructions below separate the dry tissue weight into size classes, which is the same input you enter into the estimator.

The Estimator works out how much water one animal clears from its size and the water temperature, and for most species it measures size as dry meat weight. Without your own number, you depend on a published shell-size conversion where one exists, and those come from other farms and other seasons. Meat weight changes a lot with season, spawning and food, so your own measurement is the better input.

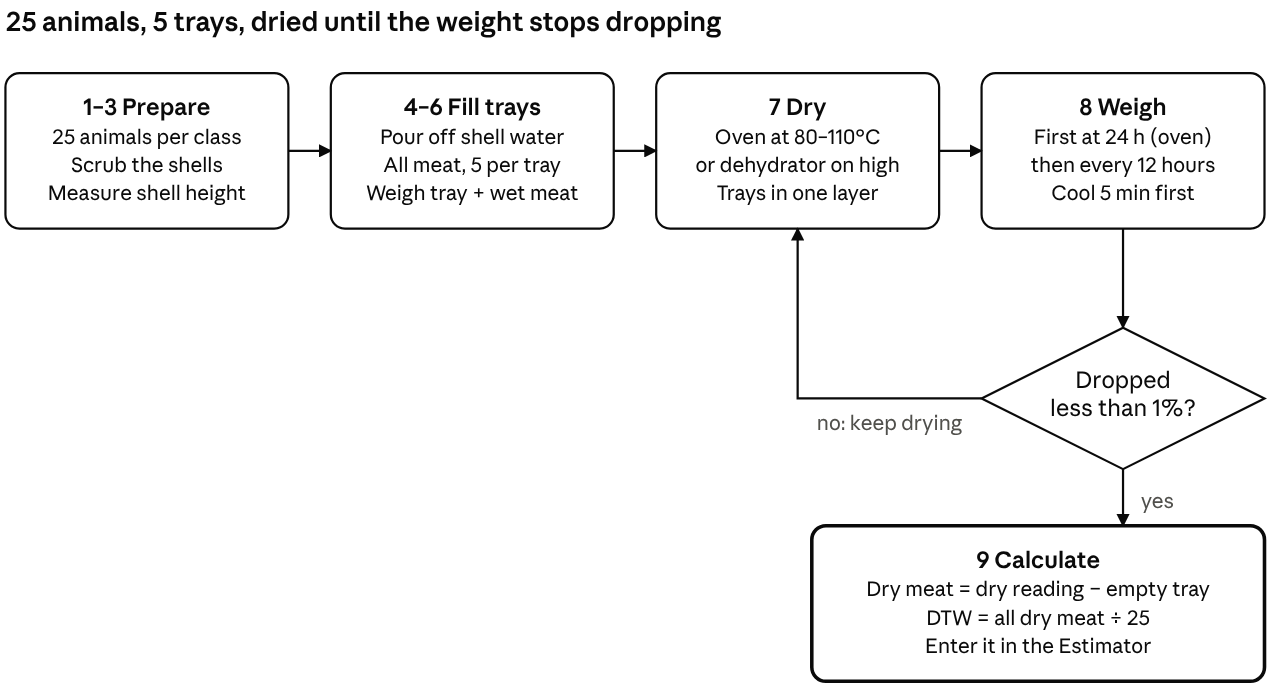
You do not need a laboratory. For most species in the Estimator, filtration rises a little more slowly than dry weight: a 10% error in dry weight moves the filtration estimate by about 4 to 10%, depending on the species. Careful kitchen work is good enough.

What you type into the Estimator, per size class	Where it comes from
Shell height (mm)	Step 3 of this protocol: the average of your measured animals
DTW (g)	Step 9 of this protocol: average dry meat weight of one animal
Number of bivalves	Your own stock records

If the Estimator hides the dry weight box for your species, it needs shell size only and you can skip this protocol. At present that is the case for blue mussel (*Mytilus edulis*).

Time needed (estimate): about an hour of hands-on work per size class, then 1 to 3 days of drying with a few minutes of weighing each day.

At a glance



Dry meat weight workflow · 5 stages, 1 drying loop

The only loop is drying: each tray goes back in until its reading drops by less than 1% between two weighings.

What you need

Everything here is household or hardware-store equipment. The one item to choose carefully is the scale: **it must be sensitive enough to read to 0.01 g.**

Item	What it is for	Notes
Digital scale reading to 0.01 g, holding at least 200 g	Every weighing	Often sold as a pocket or jewelry scale. A kitchen scale reading to 1 g or 0.1 g is not precise enough. Use the same scale for every weighing.
Oven thermometer	Checking the real temperature where the trays sit	Must read down to 70°C (160°F). Oven dials are often several degrees off.

Item	What it is for	Notes
Kitchen oven or toaster oven, or a food dehydrator	Drying the meat	Set-up check A says which to use.
Foil trays, 5 per size class	Holding the meat while it dries and is weighed	For example foil tart or pie tins, or heavy-duty foil folded into shallow trays. Wide and shallow dries fastest.
Baking sheet	Carrying the trays in and out together	Any flat metal sheet that fits your dryer.
Oyster knife or short strong knife, and a cut-resistant glove	Opening the animals safely	
Calipers, or a ruler marked in millimetres	Measuring shell height	
Small scissors and tweezers	Cutting large pieces of meat, picking out shell chips	
Stiff brush	Cleaning the outside of the shells	
Bowl	Catching the water inside the shell as you open it	
Oven mitts and tongs	Handling hot trays	
Ballpoint pen or permanent marker	Numbering the trays	
One coin, kept only for this job	Checking the scale at every session	Any coin works; it just has to be the same coin every time.
Cooler with ice packs and a damp cloth	Keeping animals alive and cool until you open them	
Airtight tin or glass jar with silica gel packets	Cooling dried trays in humid weather	Helpful, not essential.

Item	What it is for	Notes
Bucket of clean seawater from your site	Letting clams, or animals from muddy water, clear their guts (Step 2)	Only for those animals.
Data sheet (a separate PDF that comes with this protocol) and a pen	Recording every number	

Use what you have. The studies behind this protocol used laboratory ovens and balances. Kitchen equipment does the same job if it passes the two checks below: the thermometer reads 80 to 110°C beside the trays, and the scale reads your check coin the same way every time.

Set up first

Do these three checks before your first batch. Check A takes about an hour, mostly waiting; B and C take a few minutes.

A. Check your dryer's real temperature

1. Put the oven thermometer on the middle rack, where the trays will sit.
2. Set the oven to **100°C (210°F)**. Use the fan (convection) setting if there is one. Wait 30 minutes, then read the thermometer.
3. If it reads between **80 and 110°C (175 and 230°F)**, use the oven at that setting. If not, turn the dial up or down and read again after 30 minutes.
4. If your oven cannot hold that range, use a food dehydrator on its highest setting instead. Most home dehydrators top out around 70°C (160°F), so drying takes longer (Step 8).
5. Many ovens switch themselves off after 12 hours as a safety feature. Check the oven each morning and evening, and switch it back on if needed. Your oven's manual may show how to turn this off.

Mo and Neilson (1994) found that oven temperatures from 80 to 105°C changed final dry weights by 1% or less; cooler ovens only took longer. This protocol caps the range at 110°C to stay close to those tested temperatures and to avoid scorching in ovens that run hot.

B. Check your scale

1. Set the scale on a firm, level surface away from the oven, fans and open windows. Moving air changes the reading. Close the scale's lid if it has one.
2. Switch it on, wait one minute, and press zero (tare) with nothing on it.
3. Calibrate it with the weight that came with it, following its instructions.
4. Weigh your check coin and write the reading at the top of the data sheet. At the start of every weighing session, weigh the coin again. If it reads more than 0.02 g away from the first reading, calibrate again or change the battery before weighing any trays.

C. Number and weigh the empty trays

1. You need 5 trays for each size class. Give each size class a letter, and number its trays A1 to A5, B1 to B5, and so on.
2. Press the number into the foil with a ballpoint pen, or write it with a permanent marker. Number the trays before you weigh them.
3. Check each tray is clean and dry, then weigh it on its own and write the weight in the **Empty tray** row of the data sheet. Never press zero (tare) with a tray on the scale: you need the empty-tray weight later.

Method

Run Steps 1 to 9 once for each size class you will enter in the Estimator. Each size class uses 25 animals in 5 trays of 5.

Step 1. Collect 25 animals for the size class.

- Take them from at least three different bags, ropes or lines holding that size class. Take them as they come: do not pick the biggest or best-looking.
- Collect on a day when you can open them within 24 hours.
- Sample in the same season as the water temperature you will enter in the Estimator. Meat weight rises and falls with season and spawning, so repeat this protocol for each season you estimate.

Baggett et al. (2014) set 25 oysters as the minimum sample for condition index; this protocol uses the same number.

Step 2. Keep them cool, then clean the shells.

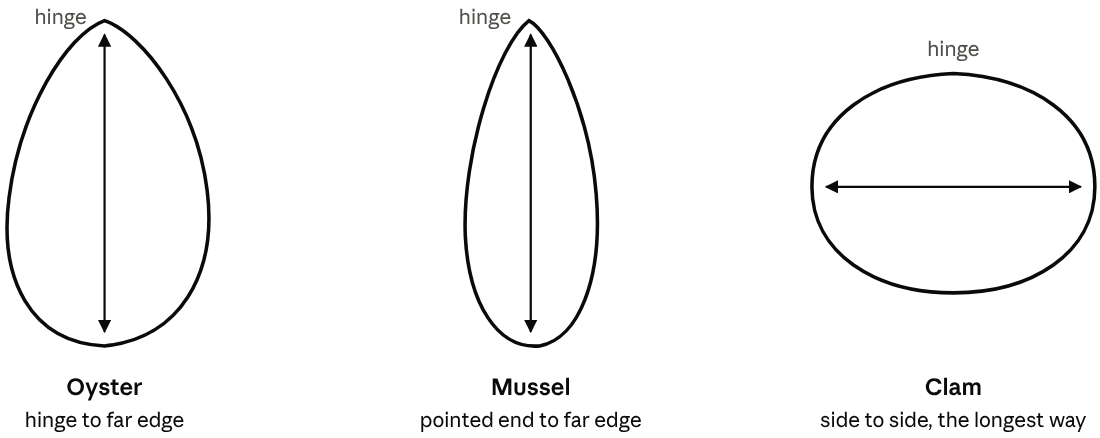
- Carry them in the cooler on ice packs, under a damp cloth. Keep them out of fresh water and melt water.
- Clams, and animals from muddy water: hold them overnight in a bucket of clean seawater from your site, kept cool and shaded, so they clear sand and mud from their guts. Mo and Neilson (1994) held their oysters in filtered seawater for a day for the same reason.
- Scrub the outside of each shell under running water. Remove barnacles, weed and mud so nothing falls into the meat when you open it.

Step 3. Measure shell height.

- Measure the longest straight line across each closed shell, to the nearest millimetre. For oysters, scallops and pearl oysters this runs from the hinge to the far edge.
- Write each height in the shell height table of the data sheet, five animals per tray.

Measure the longest straight line across the closed shell

Calipers or a ruler, to the nearest millimetre



Where to measure shell height · oyster, mussel, clam

Measure each shell closed, before you open it. For clams the longest line runs side to side, not from the hinge.

Step 4. Open each animal over the bowl and pour off the water inside the shell.

- Wear the cut-resistant glove on the hand that holds the shell.
- Oysters: push the knife into the hinge and twist. Mussels, clams, scallops and pearl oysters: slide the knife between the shells along the edge where they open.

- Cut the muscle that holds the shell shut, open the shell, and let the seawater inside run into the bowl. It is not part of the animal, and its salt would add to the dry weight. Mo and Neilson (1994) pour this water off the same way.

Step 5. Put all of the meat, and nothing else, into the tray.

- Scrape the meat off both shells, including the muscle stuck to each one, and lift it straight into its tray. Five animals go in each tray.
- Do not rinse, squeeze or blot the meat, and leave any juice that runs from it in the tray. Mo and Neilson (1994) report that fluid seeping from shucked oyster meat can be as much as half its starting weight, and that rinsing, blotting or draining causes errors in dry weight.
- If the meat lies in a heap, cut big pieces with scissors inside the tray so it spreads into one layer. If five animals will not fit in one layer, use a bigger tray.

Keep in the tray	Take out
All of the soft body: gills, mantle, muscle, gut, roe, foot and siphons	The water inside the shell (Step 4)
For scallops, everything, not just the white muscle people eat	Mussel beard (byssus threads): cut it off
Any juice that runs from the meat into the tray	Pea crabs, worms and anything else living inside the shell
	Shell chips, grit and sand
	Pearls and implanted pearl beads

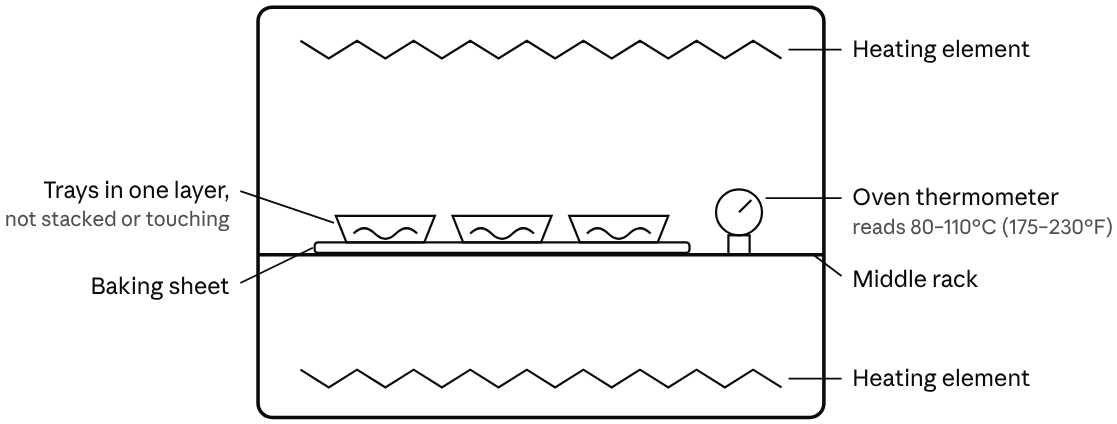
Step 6. Weigh each tray with its wet meat.

- Weigh it as soon as the tray is full and write it in the **Tray + wet meat** row. This is only a check value for Step 9.

Step 7. Dry the trays.

- Set the trays in one layer on the baking sheet, not stacked or touching, with the thermometer beside them. Slide the sheet onto the middle rack, away from the walls and heating elements.
- Oven: use the setting you found in check A. Dehydrator: use its highest setting.
- Keep the door closed except when weighing.

Trays in one layer on the middle rack, thermometer beside them

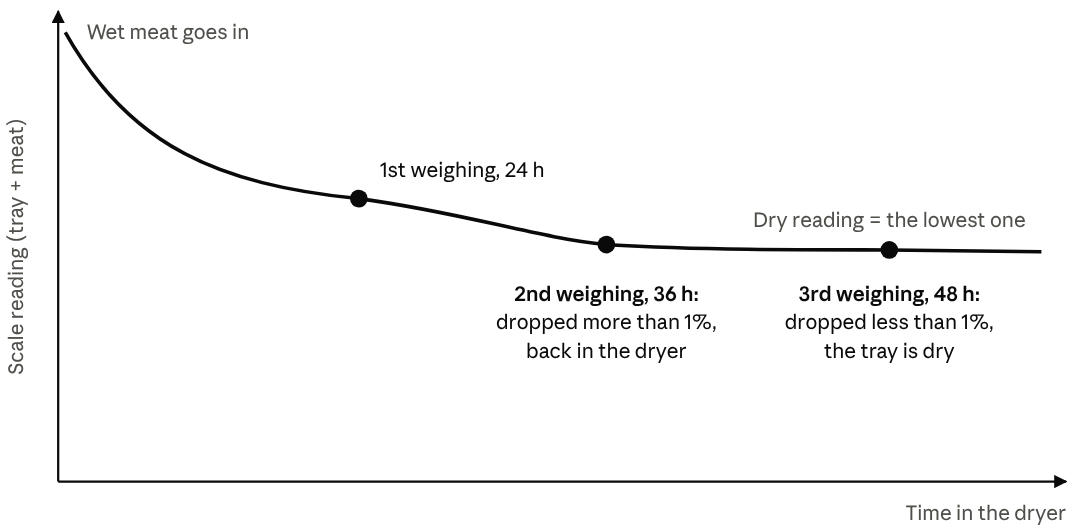


Dryer set-up · trays, rack and thermometer

The thermometer sits at tray level because that is the temperature the meat actually gets; the oven dial can be several degrees off.

Step 8. Weigh, put back, and weigh again until the weight stops dropping.

- First weighing: 24 hours after the trays go in an oven, or 48 hours in a dehydrator. After that, weigh every 12 hours, for example morning and evening.
- At each session, weigh the check coin first (check B). Take out up to 5 trays, let them cool for 5 minutes, weigh each one, write it down, and put them back. Finish within 10 minutes of taking them out, because dried meat soaks up moisture from the air. In humid weather, cool them in the airtight tin with silica gel.
- If the oven switched itself off, switch it back on and wait one hour at temperature before weighing.
- A tray is dry when its reading has dropped by **less than 1%** since the last weighing. To find 1%, move the decimal point two places left: 1% of 8.74 g is 0.087 g, so the next reading must be 8.66 g or more. A reading that goes up slightly also means the tray is dry.
- Keep drying any tray that is not dry yet. A tray's **dry reading** is the lowest reading it reached.
- Expect 1.5 to 3 days in an oven at 100°C, and longer in a dehydrator. Mo and Neilson (1994) reached constant weight in 36 hours at 100°C and 60 hours at 80°C; zu Ermgassen et al. (2021) give 36 to 72 hours at 100°C, depending on oyster size.



Drying to constant weight · example of one tray

The reading falls fast at first, then levels off. The tray is done when two weighings 12 hours apart differ by less than 1%.

Step 9. Work out the dry meat weight per animal.

- For each tray: **dry meat = dry reading − empty tray**.
- Add up the dry meat of all five trays and divide by the number of animals (25). This is the **DTW (g)** for the size class.
- Average the 25 shell heights. This is the **shell height (mm)**.
- Enter both numbers in that size class's row of the Filtration Service Estimator.

Worked example and checks

In this example, the five trays of size class B hold 33.98 g of dry meat in total, so the DTW to enter is 33.98 ÷ 25 = **1.36 g per animal**. The numbers are made up to show the arithmetic.

Size class B	B1	B2	B3	B4	B5
Animals in tray	5	5	5	5	5
Empty tray (g)	2.14	2.11	2.16	2.12	2.15
Tray + wet meat (g)	62.44	60.21	51.96	63.32	54.55
Dry reading (g)	8.71	9.40	8.05	9.88	8.62

Size class B	B1	B2	B3	B4	B5
Dry meat = dry reading − empty tray (g)	6.57	7.29	5.89	7.76	6.47
Wet meat = tray + wet meat − empty tray (g)	60.30	58.10	49.80	61.20	52.40
Dry meat as % of wet meat	10.9	12.5	11.8	12.7	12.3

If the 25 shell heights average 88 mm, the Estimator row for size class B gets **shell height 88 mm** and **DTW 1.36 g**.

How tray B1 reached its dry reading:

Weighing	Reading (g)	Drop since last (g)	1% of last reading (g)	Dry?
24 hours	9.05			
36 hours	8.74	0.31	0.09	No: dropped more than 1%
48 hours	8.71	0.03	0.087	Yes: dropped less than 1%

Checks before you enter the numbers

- The five trays should give similar "dry meat as % of wet meat" values. If one tray stands far apart from the other four, look for shell grit, a tray that is not fully dry, or a subtraction slip.
- For oysters, Mo and Neilson (1994) list published values of dry meat from about 10% to 27% of wet meat. A size class well outside that range deserves a second look.
- Dry meat per animal will differ from tray to tray because animals differ; that is why the protocol averages 25 of them.

What this number can and cannot tell you. It gives the average dry meat weight of one animal in this size class, on your farm, in this season: the DTW the Estimator asks for. It does not cover other seasons or size classes, which need their own run of this protocol. It is not a measure of how much your animals actually filter; the Estimator turns it into a best-case estimate, with the caveats listed in the Estimator itself.

Expect these mistakes (and avoid them)

Most mistakes push the dry weight up (salt, shell, damp meat) or down (lost juice, scorching). Each one below has a step that prevents it.

Mistake	What it does to your number	Do this instead
Rinsing, squeezing or blotting the meat	Body fluid is lost, so dry weight comes out low	Lift the meat straight into the tray and keep its juice (Step 5)
Leaving the shell water in	Sea salt is added, so dry weight comes out high	Pour the water off before cutting the meat free (Step 4)
Freezing, steaming or cooking animals to open them	Thawing and cooking release juices that are easily lost	Open them alive, within 24 hours of collection (Steps 1 and 4)
Picking the best-looking animals	The average comes out high	Take them as they come, from three or more bags or lines (Step 1)
Leaving in beard, pea crabs, shell chips or sand	Weight that is not meat is added	Check every tray against the keep / take out table (Step 5)
Piling the meat thick	Slow drying, with wet centres	Spread it in one layer and cut big pieces (Step 5)
A scale that reads only to 0.1 g or 1 g	Rounding errors swamp the result	Use a 0.01 g scale and check it with the coin every session (check B)
Pressing zero (tare) with the tray on the scale	The empty-tray weight is lost	Weigh empty trays on their own and write them down (check C)
An oven that is not checked	Too hot can scorch the meat; too cool dries slowly	Confirm 80 to 110°C beside the trays with the thermometer (check A)
Stopping after a fixed time	Meat may not be fully dry, so dry weight comes out high	Dry until the reading drops by less than 1% between weighings (Step 8)

Mistake	What it does to your number	Do this instead
Weighing trays hot	Warm air rising off the tray makes the scale read low, and heat can harm the scale	Let trays cool for 5 minutes first (Step 8)
Leaving dried trays out	Dried meat soaks up moisture from the air, so dry weight comes out high	Weigh within 10 minutes; in humid weather, cool trays in the silica gel tin (Step 8)
Changing the scale, dryer or steps between batches	Results can no longer be compared	Keep everything the same each time, and note any change on the data sheet

Safety and hygiene

The knife is the main hazard, and drying shellfish smells strongly, so plan where you will dry.

- Wear a cut-resistant glove on the hand holding the shell, and push the knife away from your body.
- Use oven mitts or tongs for hot trays and the baking sheet.
- If you can, use a toaster oven or dehydrator kept only for this job, in a well-ventilated shed or porch. If you use the household oven, keep the trays on the baking sheet and wipe the oven out afterwards.
- Follow the oven's or dehydrator's safety instructions for long running times, and keep the area around it clear.
- The dried meat is not food: throw it away after the final weighing. Wash the knife, bowl and any trays you will reuse in hot soapy water before they go near food, and wash your hands after handling raw shellfish.
- Keep children and pets away from knives, hot surfaces and drying trays.

References

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TNC Filtration Service Estimator and its clearance-rate workbook (Clearance_rate_estimation_tool_TNC.xlsx)